MILITARY SPECIFICATION

DEVICE, RESTRAINT HARNESS TAKE-UP, INERTIA-LOCKING POWERED-RETRACTING: FOR USE IN SIIIS-3 EJECTION SEAT

This specification has been approved by the Naval Air System Command, Department of the Navy

The complete requirements for procuring the Restraint Harness Take-up Device described herein shall consist of this document and the issue in effect of MIL-D-81514.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for SIIIS-3 Inertia-Locking Powered-Retracting, Restraint Harness Take-up Device (PIRD).

2. APPLICABLE DOCUMENTS

2.1 The applicable documents shall be in accordance with the general specification.

3. REQUIREMENTS

3.5.2.3.2 Firing mechanism - The gas generator shall incorporate a gas actuated, percussion type, firing mechanism.

3.5.2.3.3 Dimensions - All dimensions shall be the minimum practical consistent within the envelope drawing of Figure 2.

3.5.2.4 Restraint harness attachments - Webbing for the restraint harness take-up strap attachment shall meet the applicable requirements of MIL-W-25361. The webbing shall be securely attached to the stowage/take-up mechanism in such a manner that field replacement can be made readily by one maintenance man without disassembly of the device or removal of the device from the seat. An end fitting shall be attached to the restraint harness take-up strap. The design of the end fitting shall be subject to the approval of the procuring activity. The webbing shall be protected at the point of attachment to prevent damage by snagging or rubbing and be protected from overlapping during operation.

3.5.2.4.1 Strap dimensions. The width of each strap shall be 3/4 of an inch. The Strap length shall be a minimum of 36 inches. The strap thickness shall be the minimum practical consistent with the requirements of this specification.

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3.5.2.4.2 Strength. The minimum breaking strength of each strap shall be 2500 lbs.

3.5.2.4.3 Take-up length. The take-up length of each strap shall be 11 inches.

3.5.2.6 Manual control. Applicable. The supplier shall provide the control handle. (See Figure 1).

3.6.1.3 Stowage/ take-up mechanism. As installed in the aircraft seat, in the manner intended for service use, it shall retract the restraint harness take-up strap automatically and shall produce a strap tension of not less than 5 pounds or more than 15 pounds throughout the range of extension.

3.7 Weight. The weight of the PIRD shall not exceed 4.0 pounds (excluding gas generator and associated tubing).

3.8 Mounting provisions. The PIRD shall be mounted in accordance with Figure 2.

3.12.1 Demonstrated reliability. The PIRD shall demonstrate a reliability of at least 0.94 at 90 percent confidence, based upon service release testing.

3.12.2 Reliability growth curves. A reliability growth curve shall be developed to assess the PIRD reliability at 90 percent confidence.

3.12.3 Reliability demonstration ground rules. The definitions and requirements below shall apply for the purpose of demonstrating the PIRD reliability requirements of this document.

3.12.3.1 Relevant failure. All failures are relevant unless determined by the procuring activity to be caused by a condition external to the PIRD.

a. Design/workmanship failures. Failures due to design deficiencies or poor workmanship of either the PIRD or component parts shall be classified relevant.

b. Component part failure. Failures due to defective component parts shall be classified as relevant failures. In the event that several component parts of the same type fail during test, each one shall be considered a separate relevant failure, unless it can be shown that one failure caused one or more of the other.

c. Intermittent failures. The first occurrence of an intermittent failure shall be counted as a relevant failure.

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3.12.3.2 Non-relevant failure. Although non-relevant failures are not used for reliability demonstration calculations, all failures shall be analyzed and reported. The following type of failures is to be classified non-relevant.

a. Failures directly attributable to improper installation in the test chamber.

b. Failures of test instrumentation or monitoring equipment.

c. Failures resulting from test operator error in setting up, or in testing of the equipment.

d. Dependent failures (at least one relevant failure shall be counted when a dependent failure is counted).

e. Failures attributable to an error in the test procudure, when the failure mechanism induced by the test procedure is repeated.

f. The second (and any subsequent) occurrence of the same intermittent failure in the same unit.

g. Failures occurring during repair verification and set-up time.

3.12.3.3 Reclassification. A failure, classified as relevant, may be reclassified to non-relevant to the current specific design, provided that all of the following conditions are met:

a. Corrective action (an equipment design, part, or production process change) has been implemented into the current specific design to eliminate the cause of failure and prevent recurrence.

b. Sufficient test data has been accumulated to indicate the corrective action into the current specific design is effective in eliminating the failure mode and failure cause.

c. Approval of the required corrective action by the procuring activity is obtained for reclassification of the future.

3.19 Life. The PIRD shall have a minimum shelf life of two years, and a service life of five years (seven years total).

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with the general specification.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the general specification.

6. NOTES

6.1 Intended use. The PIRD is intended for use as a shoulder restraint take-up mechanism on an SIIIS-3 Ejection Seat.

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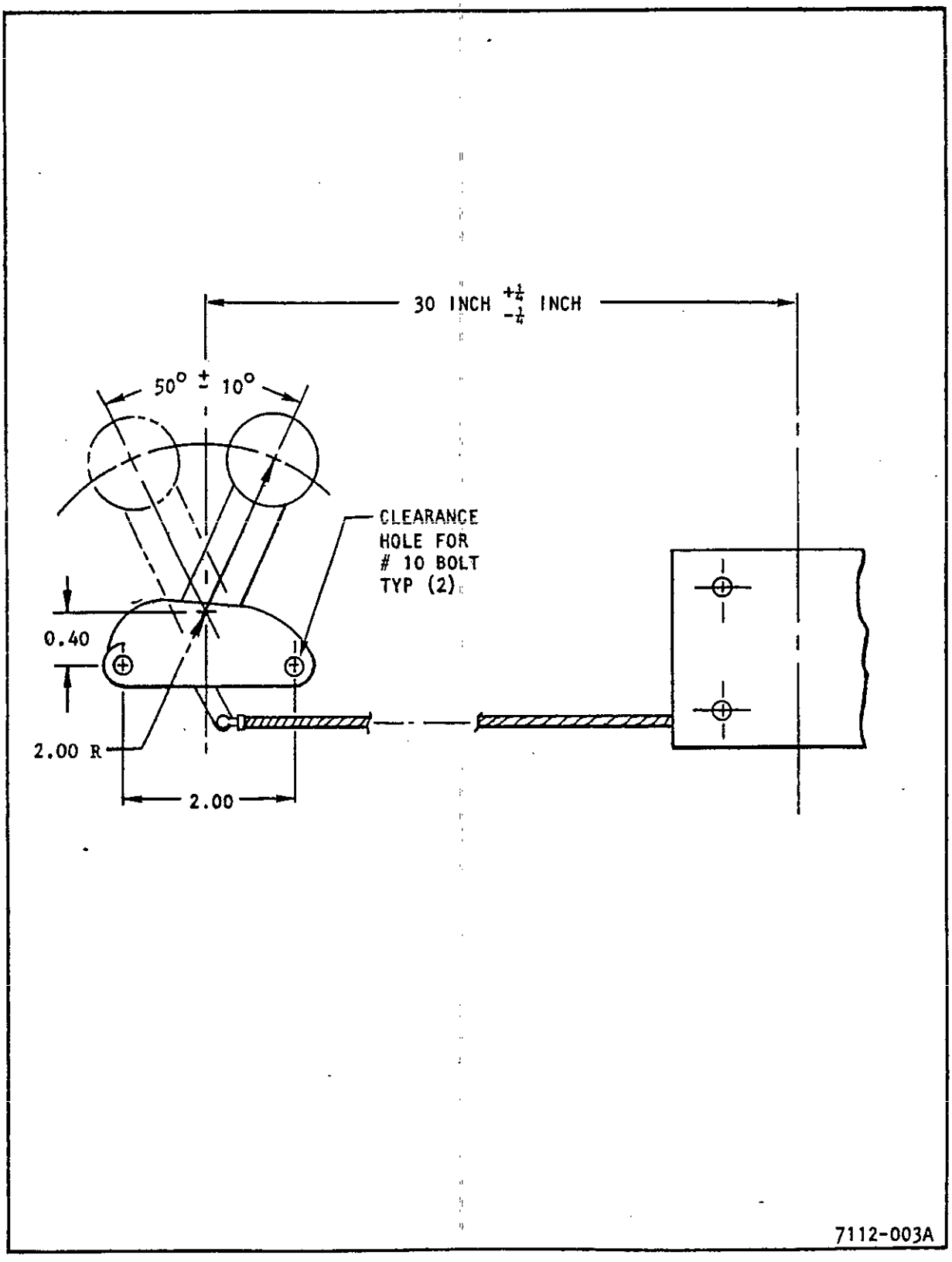


Figure 1. Manual control dimensions

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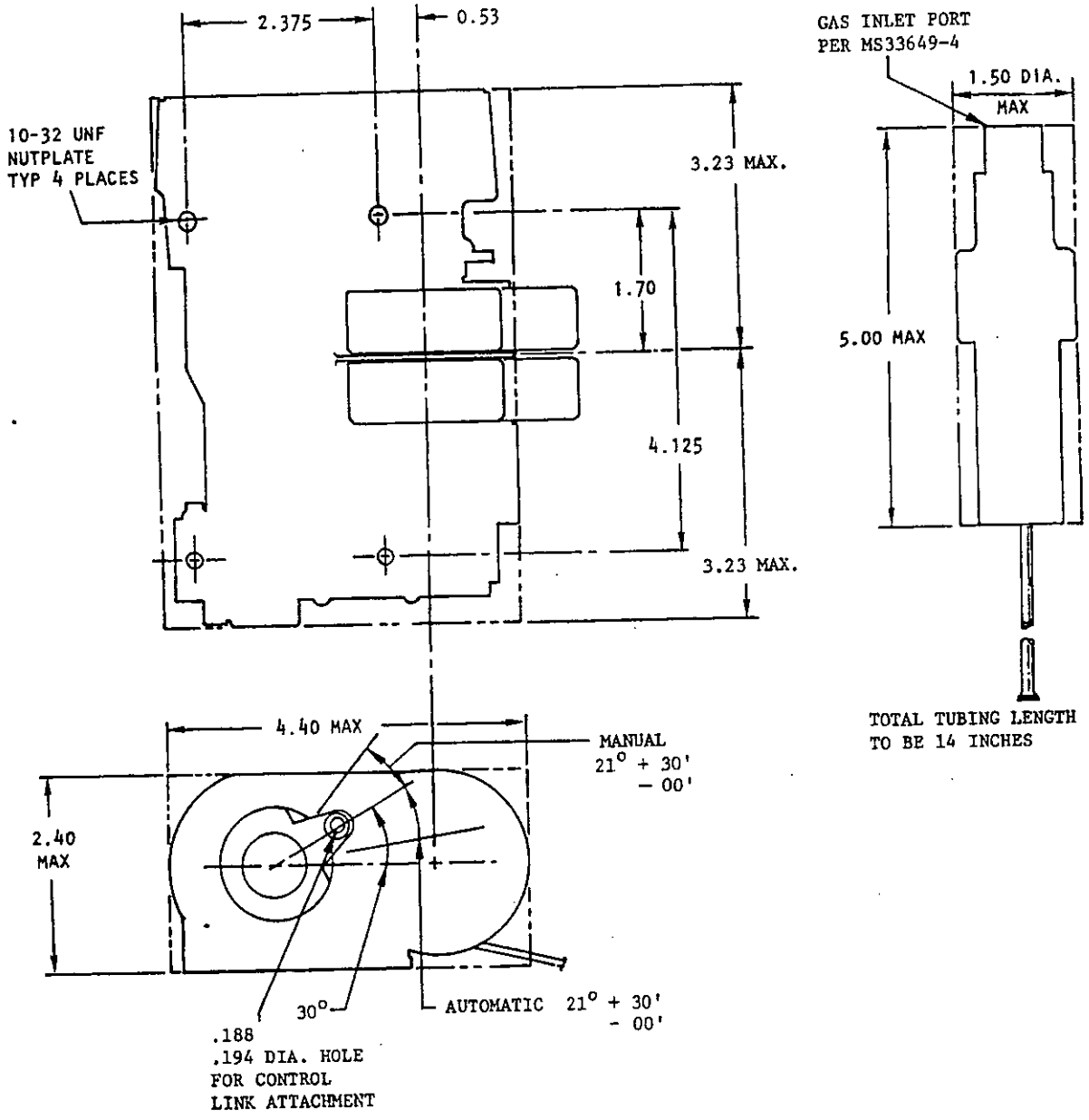


Figure 2. Mounting and envelope dimensions

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-D-81514/1A(as) DEVICE, RESTRAINT HARNESS TAKE-UP, INERTIA-LOCKING POWERED-RETRACTING: FOR USE IN CITIS-3 EJECTION SEAT

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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